UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 (5 marks)

The Von Neumann Architecture was first described in a seminal First Draft of a Report on the EDVAC. The document describes a design architecture of a general-purpose electronic computer that serves as the basis for almost all of modern computers.

a) List TWO (2) main components of a Von Neumann Architecture.

b) Von Neumann computers are general purpose computers. Define the term general-purpose computers.

c) Name ONE (1) of the earliest computers built based on the Von Neumann Architecture.

QUESTION A2 (5 marks)

The integrated circuit is an important milestone that marks the achievement in the history of computer development.

a) In which computer generation that the integrated circuil was first introduced?

b) State the name of the inventor of the integrated circuit.

c) What are the THREE (3) main components of integrated circuits?

QUESTION A3 (5 marks)

a) \Mat is a bit?

b) Given a system of vacuum tubes where each tube represents a single bit with two states: ON=1 and OFF=O. Identify the number of vacuum tubes needed to store the value of gro. Show your workflow to justify your answer.

   i. Binary system.

   ii. Octal system.

QUESTION A4 (5 marks)

Given the following Figure 1.

a) Label the addressing technique illustrated in Figure 1.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

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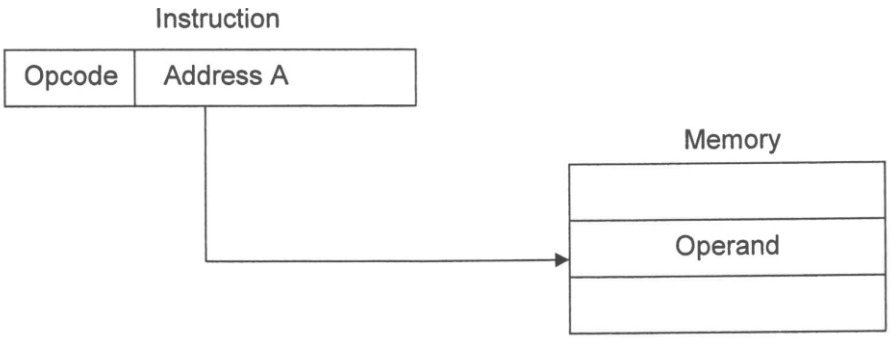


Figure 1

QUESTION A5 (10 marks)

a) Discuss the THREE (3) input-output techniques available in the computer system.

b) Universal Serial Bus (USB) has a "hot connection" to the devices. Briefly explain what is meant by "hot connection."

c) Why is there a need for the I/O module to be the mediator between the CPU and devices?

QUESTION A6 (10 marks)

a) Differentiate SRAM and DRAM from the perspective of the following characteristics,

i. The way the bits are stored.

ii. Speed.

iii. Location it is being used.

b) Why does DRAM need refreshing?

c) Comparing DRAM and SRAM, which memory would be placed higher in the memory hierarchy? Give your justification to support your answer.

QUESTION A7 (10 marks)

a) List FOUR (4) classification of multiple processor architecture.

b) Symetric Multi Processor (SMP), Non Uniform Memory Access (NUMA) and Cluster are part of the X multiple processor architecture. Identify X?

c) Differentiate the multiple processor architecture for SMP and Cluster. In your opinion, which architecture is easier to manage and control by giving appropriate justification.

QUESTION A8 (10 marks)

a) Identify the correct concept of computer organization or architecture in the following scenario.

Scenario

Concept

Deals with the logical aspects of the system, including instruction sets, data formats, and addressing modes.

Defines the functionality, capabilities,

and programming model.
Concerned with how the computer will execute the tasks and implement the functions.
Involves the hardware layout, circuits, control signals, memory access, and timing.

b) Describe THREE (3) CPU registers involved in data processing, such as arithmetic operations.

QUESTION A9 (10 marks)

a) Draw a complete machine cycle by giving the exact location of each phase.

b) Briefly, describe each phase in the machine cycle.

QUESTION A10 (10 marks)

Each generation is well known for its technology and machine invented. Complete the following Table 1 with appropriate answers.

Generation	Technology	Machine	Inventor
Zero generation	Mechanical computer	Difference Machine	_____(i)_____
		_____(ii)_____ (ABC)	John V. Anatasoft and Clifford Berry
		_____(iii)_____	Gottfried von Leibniz
First generation	_____(iv)_____	_____(v)_____ (ENIAC)	John W. Mauchly and J. Presper Eckert
		_____(vi)_____ (EDSAC)	Maurice Wilkes
Second Generation	_____(vii)_____	IBM 7090	
Third Generation	Integrated Circuit	_____(viii)_____	Jack Kilby
Fourth Generation	_____(ix)_____		
Fifth Generation	_____(x)_____		

Table 1

PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 (20 marks)

Question 1 is based on Figure 2 and Table 2.

Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 2 using one-, two- and three-address instructions. The instructions available for use are provided in Table 2. Note that Register A, B, C, D, E and F cannot be modified and Register X, Y and Z can be used as temporary register. Final answer must be stored in Register X.

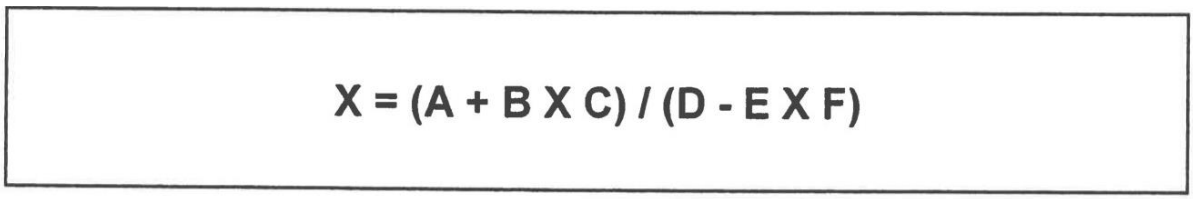


Figure 2

Table 2 : Machine Instruction

i Nia dress		**address**
LOAD M	MOVE (X e Y)	MOVE (X e Y)
STORE M	ADD (X k.. X + Y)	ADD (X e Y + Z)
ADD M	SUB (X e X - Y)	SUB (X e Y-Z)
SUB M	MUL (X e X x Y)	MUL (X e Y x Z)
MUL M	DIV (X e x / Y)	DIV (X e Y / Z)
DIV M		